

ATIYAH'S MINKOWSKI SPACE CONJECTURE FAILS FOR EVERY $n \geq 3$

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ABSTRACT. Atiyah's Minkowski-space version of the configuration-of-points construction assigns to an admissible marked configuration of n worldlines a collection of n binary forms of degree $n - 1$, whose roots are the ordered retarded celestial directions. He conjectured that these forms are always linearly independent.

We disprove this conjecture for every $n \geq 3$. For $n = 3$, an explicit planar one-parameter family yields a real coefficient determinant with exactly one simple zero in a specified interval. At this parameter, all six ordered celestial roots are distinct and the coefficient matrix has rank exactly two. A null-translation construction then multiplies the first three forms by a common factor and produces counterexamples for every $n > 3$. Consequently, within the class of complete pairwise disjoint timelike affine lines, universal independence holds at $n = 2$ and fails for every $n \geq 3$; for each of the counterexamples, the normalized Atiyah–Sutcliffe determinant is defined and vanishes.

1. INTRODUCTION

1.1. Atiyah's Minkowski-space conjecture. The configuration-of-points problem rests on a precise dimension match. Let

$$W_n = \text{Sym}^{n-1}((\mathbb{C}^2)^*), \quad \dim_{\mathbb{C}} W_n = n.$$

For each label i , Atiyah's construction multiplies the $n - 1$ linear factors determined by the ordered directions from i to the remaining labels and thereby produces a binary form $\beta_i \in W_n$. Thus one obtains exactly n vectors in an n -dimensional vector space. Their linear independence is the nondegeneracy condition that turns the projective classes $[\beta_1], \dots, [\beta_n]$ into an ordered projective frame. After fixing a Hermitian metric, one may choose unit representatives and place them in the columns of an invertible matrix. The unitary polar factor of that matrix determines an ordered orthogonal frame, hence a complete flag. Changing the unit representatives only changes the phases of the columns and therefore leaves the corresponding point of $U(n)/T^n$ unchanged. Berry and Robbins's transported-spin construction motivated the configuration-to-flag problem [6]. Atiyah then formulated the associated equivariant flag-map problem, gave an explicit solution of its bare existence version, and proposed the binary-form ansatz used here [1, 2].

Atiyah also formulated a retarded Minkowski version of the same ansatz. He introduced the construction in [2, §6] and stated the conjecture explicitly in [3, Conjecture 1.6.1]. In the notation used here, Atiyah's original conjecture is the following.

Conjecture 1.1 (Atiyah's Minkowski space conjecture). *Let $n \geq 2$. Let $\xi_1, \dots, \xi_n \subset \mathbb{M}^{3,1}$ be n nonintersecting worldlines, and let $x_1, \dots, x_n$ be n distinct events with $x_i \in \xi_i$ for every i . For each $i \neq j$, let u_{ij} be the point in the celestial sphere of x_i at which the past light cone at x_i meets ξ_j . Identify these celestial spheres by parallel translation and form the polynomials β_i from the roots u_{ij} , $j \neq i$. Then $\beta_1, \dots, \beta_n$ are linearly independent over $\mathbb{C}$.*

The conjecture is stated in the context of the preceding retarded construction and therefore presupposes that the indicated celestial points are defined. For the proofs below, we work in a smaller, geometrically well-posed subclass. Namely, let $\xi_1, \dots, \xi_n$ be pairwise disjoint future-directed

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timelike worldlines, choose a marked event $x_i \in \xi_i$ on each worldline, and suppose that, for every $i \neq j$, the past light cone of x_i meets ξ_j in a unique point y_{ij} . We call such marked data *admissible*. Because Minkowski space is flat, parallel translation identifies the celestial spheres based at the different marked events with one common sky, exactly as in Atiyah's formulation [3, §1.6]. The past-null ray

$$u_{ij} = [y_{ij} - x_i]_+, \quad [v]_+ := \{\lambda v : \lambda > 0\},$$

then determines, after one common projective identification of this sky with $\mathbb{P}^1(\mathbb{C})$, a linear factor ℓ_{ij} , and the i th form is

$$\beta_i = \prod_{j \neq i} \ell_{ij} \in W_n.$$

Section 2 gives the causal construction, lift conventions, and coordinate invariance in full detail.

The main theorem below refutes Conjecture 1.1. Indeed, our counterexample consists of complete, pairwise disjoint, future-directed timelike affine lines, and every ordered retarded intersection exists uniquely. It therefore satisfies Atiyah's original geometric input while the associated polynomials are linearly dependent. We construct the counterexamples by restricting attention to the admissible timelike subclass, which serves only to make every retarded datum unambiguous and to show that the failure already occurs for uniform subluminal motion; of course a counterexample in this subclass is automatically a counterexample to the original conjecture.

Atiyah singled out the case $n = 3$ as a prove-or-disprove challenge and identified uniform motion as a natural version of the question [3, §1.6]. The purpose of this paper is to give a negative answer in that very class and then to propagate the failure to every particle number $n \geq 3$.

1.2. Previous results and the remaining Minkowski case. The literature separates naturally into the theory concerning the flag-map problem, positive results for the Euclidean and hyperbolic branches, and extensions of the polynomial formalism. We review these developments before isolating the uniform-motion Minkowski problem left open by Atiyah.

Foundations and the flag-map problem. The transported-spin construction of Berry and Robbins supplied the physical and geometric motivation [6]. Atiyah formulated the resulting equivariant flag-map problem, gave an explicit solution of its bare existence version, and proposed the more rigid binary-form construction considered here in [1] and [2]. Atiyah and Bielawski subsequently proved the existence of smooth W -equivariant flag maps for arbitrary compact Lie groups by using Nahm's equations [4]. That existence theorem does not, however, prove the nondegeneracy of Atiyah's specific binary forms. Atiyah introduced the forms and their coefficient determinant in [1] and [2]. Atiyah and Sutcliffe then developed the scale-invariant normalized determinant systematically, formulated the stronger Conjectures II and III, supplied extensive numerical evidence, and computed the associated energy-minimizing configurations for up to 32 particles [5].

Positive results in the Euclidean branch. The Euclidean linear-independence conjecture is elementary for $n \leq 3$, and Atiyah also proved it for collinear configurations [1, 2]. Eastwood and Norbury proved it for every configuration of four points [10]. Đoković proved the conjecture for two special types of configurations, obtaining the stronger Atiyah–Sutcliffe conclusion for one of them [8]; he also treated the nonplanar family in which some points lie on a line and the remaining points form a regular polygon in a perpendicular plane whose centroid lies on that line [9]. For almost-collinear configurations, Svrtan and Urbiha proposed symmetric-function conjectures implying Conjectures II and III and carried out computations through $n = 6$; for $n = 4$ they also verified Conjectures II and III for several infinite families of tetrahedra [23]. Their later work verified Conjectures II and III for parallelograms, cyclic quadrilaterals, and further infinite tetrahedral families, proposed stronger four-point statements, and carried the multi-Schur computations through $n = 9$ [24]. Mazur and Petrenko proved Conjecture II for regular polygons and convex quadrilaterals and

Conjecture III for inscribed quadrilaterals [22]. Bou Khuzam and Johnson gave a computer-aided proof of Conjectures II and III for all Euclidean four-point configurations [7], while Malkoun later gave a new proof of four-point linear independence by showing that the associated Gram matrix is positive definite [17].

Hyperbolic, Lie-theoretic, and topological extensions. For four points in hyperbolic space, Malkoun proved linear independence for noncoplanar configurations and for configurations in which one point lies in the hyperbolic convex hull of the other three [13]. In the coplanar case, he proved the stronger bound $|D| \geq 1$ for convex quadrilaterals and also obtained the nonvanishing result attributed there to Y. Zhang and J. Ma for nonconvex quadrilaterals [16].

Beyond the unitary problem, Malkoun constructed an explicit symplectic candidate and proved its linear-independence conjecture for $n = 2$ [12]; he then gave analogous explicit constructions for the remaining classical groups [14]. He also introduced a multivariable extension of the polynomial problem [15], proved that the $U(2m)$ conjectures imply the corresponding $Sp(m)$ conjectures through root-system domination [18], formulated a weight-theoretic rank conjecture [19], and derived a global expansion formula for the Atiyah–Sutcliffe determinant [20]. His later graph extension replaces the complete graph by an arbitrary finite simple graph; for the complete graph, its first two conjectures specialize to Atiyah–Sutcliffe Conjectures I and II [21]. Finally, Guerra and Salvatore showed that suitable strong complex and planar real forms of the Euclidean conjectures would produce, respectively, E_3 - and E_2 -algebra structures on disjoint unions of unordered full flag manifolds [11].

The remaining uniform-motion problem. The Euclidean and hyperbolic branches impose reciprocity relations on oppositely ordered roots; general retarded worldlines do not. Atiyah’s 2001 cyclic construction produced dependent forms along straight spatial tracks, but the required velocities were nonuniform. He therefore asked whether the example could be modified to use straight worldlines in Minkowski space, or whether linear independence might instead hold for all such worldlines [2, §6, p. 1387]. In the 2010 lectures he recalled that the earlier proposed realization involved faster-than-light motion and again singled out uniform motion with $n = 3$ as a prove-or-disprove challenge [3, §1.6]. To the author’s knowledge, the literature cited above contains no published resolution of the uniformly moving timelike problem before the present work. The results below answer it negatively. They do not contradict the Euclidean or hyperbolic results just reviewed: the counterexample is neither static nor formed by inertial rays issuing from a common event.

1.3. The explicit family and the main results.

1.3.1. A Family of Three Worldlines. The counterexample is planar. We work in $\mathbb{M}^{2,1} = \{z = 0\} \subset \mathbb{M}^{3,1}$ with metric $-dt^2 + dx^2 + dy^2$. This inclusion preserves the causal relations and the retarded intersections used below. Consequently, a counterexample in $\mathbb{M}^{2,1}$ is already a counterexample in $\mathbb{M}^{3,1}$.

For a real parameter T , consider the complete affine worldlines

$$\xi_1(r) = (r, 0, 0), \tag{1}$$

$$\xi_2(r) = \left(-\frac{2}{3} + r, -\frac{1}{4} + \frac{r}{2}, \frac{1}{5} - \frac{r}{10}\right), \tag{2}$$

$$\xi_3(r) = \left(T + r, 1 + \frac{4r}{5}, \frac{1}{8} - \frac{r}{5}\right), \tag{3}$$

with marked events $x_i = \xi_i(0)$. The direction vectors and spatial tracks are independent of T ; only the temporal placement of the third line changes. Set

$$I = \left[\frac{3}{4}, \frac{9}{10}\right], \quad T_- = \frac{82950}{10^5}, \quad T_+ = \frac{82951}{10^5}, \quad J = [T_-, T_+].$$

1.3.2. Main Theorems.

Theorem 1.2 (The three-worldline counterexample). *There is a unique parameter $T_* \in J$ for which the three Atiyah quadratics associated with (1)–(3) are linearly dependent. For every $T \in I$, the three parametrized lines are complete and pairwise disjoint, their tangent vectors are future directed and timelike, and each ordered pair has a unique retarded intersection. At $T = T_*$:*

- (i) *the real coefficient determinant has a simple zero;*
- (ii) *the six ordered celestial roots are pairwise distinct; and*
- (iii) *the coefficient matrix has rank exactly two.*

Consequently, the associated real coefficient-matrix path meets the smooth rank-two stratum of the determinantal hypersurface $\{\det = 0\} \subset M_3(\mathbb{R})$ transversely. In particular, the timelike formulation above, and hence Atiyah's printed Minkowski-space conjecture, is false already for three observers moving uniformly with strictly subluminal velocities.

A counterexample for $n = 3$ refutes Conjecture 1.1 as a universal statement, but it does not by itself refute the assertion at each prescribed $N > 3$: adjoining worldlines changes both the number of forms and their degree. The next theorem supplies a separate geometric construction at every particle number.

Theorem 1.3 (Counterexamples for every $N \geq 3$). *For every integer $N \geq 3$, there are N pairwise disjoint complete affine worldlines in $\mathbb{M}^{2,1} \subset \mathbb{M}^{3,1}$, with future-directed timelike direction vectors and one marked event on each line, such that every ordered pair has a unique retarded intersection and the associated N binary forms of degree $N - 1$ are linearly dependent.*

Remark 1.4. The two theorems give more than the failure of a single universally quantified assertion. Corollary 7.3 shows that, within the class of complete pairwise disjoint timelike affine lines, universal independence holds exactly for $n = 2$. Corollary 7.2 shows that the normalized Atiyah–Sutcliffe determinant is defined and vanishes on every configuration constructed here. The three-worldline core has six distinct roots; the stabilization for $N > 3$ intentionally introduces repeated common roots. Atiyah's formulation starts with nonintersecting worldlines; marked events chosen on different worldlines are therefore automatically distinct. Neither the construction nor the printed conjecture requires all ordered celestial roots to be pairwise distinct; see [2, §6] and [3, Conjecture 1.6.1]. Requiring all ordered roots to remain distinct for every N would define a stronger problem.

Remark 1.5. No common-time or spacelike-slice condition appears in Atiyah's printed Conjecture 1.6.1 [3], so the marked events used here are legitimate. They are not simultaneous in any inertial frame; imposing synchronization would define a different, stronger problem. Section 7 records this and the other boundaries of the conclusion precisely.

1.4. Main idea of the construction. The proof of the three-line theorem is a real determinantal wall crossing. For each $T \in I$, Proposition 3.1 gives six admissible ordered observation–emitter pairs. Lemma 2.1 solves each retarded null equation explicitly, and the planar Cayley coordinate of Section 2 represents the six celestial roots by nonzero real lifts

$$H_{ij}(T) \in \mathbb{R}^2 \setminus \{0\}, \quad i \neq j.$$

The two roots seen by observer i determine the coefficient row of one binary quadratic. Lemma 4.1 then converts the determinant of the three coefficient rows into the six-bracket expression

$$\begin{aligned} \Delta_H(T) = & [H_{12}, H_{23}][H_{13}, H_{32}][H_{21}, H_{31}] \\ & + [H_{12}, H_{31}][H_{13}, H_{21}][H_{23}, H_{32}]. \end{aligned} \tag{4}$$

This identity is the algebraic bridge between the six retarded directions and linear dependence of the three forms.

The exact formulas in Section 4 reduce every entry of (4) to rational expressions in the three variable radicals $\sqrt{P_1(T)}$, $\sqrt{P_2(T)}$, $\sqrt{P_3(T)}$ and the three fixed radicals $\sqrt{11}$, $\sqrt{41}$, $\sqrt{65}$. Proposition 5.1, proved in Appendix A by exact interval estimates, gives

$$\Delta_H(T_-) > 0 > \Delta_H(T_+), \quad \Delta'_H(T) < 0 \quad (T \in J). \quad (5)$$

The endpoint signs give a zero, and the derivative bound makes it unique in J and simple. The same proposition gives the fifteen nonvanishing inequalities

$$[H_{ij}(T), H_{kl}(T)] \neq 0 \quad (T \in J, (i, j) \neq (k, l)). \quad (6)$$

Thus the six roots remain pairwise distinct at the crossing. Since the coefficient matrix is singular but no two quadratic rows are proportional, its rank is exactly two. At a rank-two matrix the differential of the determinant is nonzero; the additional inequality $\Delta'_H(T_*) \neq 0$ in (5) therefore gives transversality to the smooth rank-two stratum of the real determinantal hypersurface. In particular, the dependence is not produced by a collision, a lightlike limit, a repeated root, or a higher-order contact: varying only the time offset of the third line moves the retarded data through a genuine rank-two wall.

The proof for $N > 3$ uses a different mechanism. At $T = T_*$, choose a nonzero relation

$$\lambda_1 \beta_1^{(3)} + \lambda_2 \beta_2^{(3)} + \lambda_3 \beta_3^{(3)} = 0. \quad (7)$$

Section 6 constructs a Lorentzian two-plane $\Pi = \text{span}\{U_*, \omega\}$, where U_* is future timelike and ω is past null, and adjoins pairwise disjoint lines

$$\xi_k(r) = rU_* + c_k\omega, \quad 4 \leq k \leq N.$$

For each old observer $i \leq 3$ and each new emitter $k > 3$, the retarded displacement is a positive multiple of the same past-null vector ω . Hence every new emitter contributes the same factor L_* to each of the first three forms, and

$$\beta_i^{(N)} = L_*^{N-3} \beta_i^{(3)}, \quad i = 1, 2, 3. \quad (8)$$

Multiplying (7) by L_*^{N-3} gives a nontrivial relation among the N forms. This common-factor identity is therefore the geometric bridge from the nondegenerate three-line core to counterexamples at every particle number.

1.5. Notation and organization. The first index in u_{ij} is always the observer and the second the emitter. Thus x_i is the marked observation event, $y_{ij} \in \xi_j$ is the retarded emission event, and $q_{ij} = x_i - y_{ij}$ is future null. We reserve T for the one-parameter three-line family, T_* for its unique crossing in J , and N for the particle number in the stabilization.

Section 2 develops the retarded celestial forms and fixes a single projective coordinate on the planar common sky. Section 3 verifies the timelike and pairwise-disjoint nature of the explicit family. Section 4 proves the six-bracket identity and derives the six exact root lifts. Section 5 proves Theorem 1.2 from Proposition 5.1. Section 6 proves Theorem 1.3 by the null-translation mechanism. Section 7 extracts the projective balance, normalized determinant, dichotomy, and exact scope. Appendix A gives the exact rational interval proof of Proposition 5.1.

2. RETARDED CELESTIAL FORMS

This section makes the construction in Conjecture 1.1 precise and proves the two causal facts needed later: a complete timelike affine emitter has a unique retarded point, and reciprocal retarded rays of two disjoint affine lines are distinct.

2.1. The common sky and the binary forms. We first fix the causal and projective conventions. Let

$$\mathbb{M}^{3,1} = (\mathbb{R}^4, \eta), \quad \eta = -dt^2 + dx^2 + dy^2 + dz^2,$$

and write $\langle \cdot, \cdot \rangle$ for the corresponding bilinear form. A nonzero vector is timelike, null, or spacelike according as its squared norm is negative, zero, or positive; timelike and null vectors are causal. A causal vector is future directed when its time component is positive and past directed when its time component is negative. The incoming celestial sphere is the projectivized past null cone

$$\mathcal{S}^- = \{v \neq 0 : \langle v, v \rangle = 0, v^0 < 0\} / \mathbb{R}_{>0}. \quad (9)$$

Flat parallel translation identifies the null cones based at different events with this single common sky, which is precisely the identification used in Atiyah's Minkowski construction [3, §1.6].

By a future-directed timelike worldline we mean the image of a C^1 embedding whose tangent is everywhere future directed and timelike. Let $\xi_1, \dots, \xi_n$ be pairwise disjoint such worldlines and choose marked events $x_i \in \xi_i$. We call these data *admissible* if, for every $i \neq j$, the past light cone of x_i meets ξ_j in exactly one point y_{ij} . This condition makes Atiyah's retarded construction single-valued; it is automatic for the complete timelike affine lines used below. With $[v]_+ := \{\lambda v : \lambda > 0\}$, the incoming celestial root is

$$u_{ij} = [y_{ij} - x_i]_+ \in \mathcal{S}^-. \quad (10)$$

Fix one projective identification $\Phi : \mathcal{S}^- \rightarrow \mathbb{P}^1(\mathbb{C})$. If $h_{ij} = (a_{ij}, b_{ij}) \in \mathbb{C}^2 \setminus \{0\}$ lifts $\Phi(u_{ij})$, set

$$\ell_{ij}(Z_0, Z_1) = b_{ij}Z_0 - a_{ij}Z_1, \quad \beta_i = \prod_{j \neq i} \ell_{ij} \in \text{Sym}^{n-1}((\mathbb{C}^2)^*). \quad (11)$$

Changing a lift rescales the corresponding form β_i by a nonzero scalar. A common orientation-preserving projective change of coordinate acts on every form through the same invertible symmetric-power transformation. An orientation-reversing change is the composition of such a transformation with simultaneous complex conjugation. Since both operations are bijective on the coefficient space and are applied simultaneously to all forms, they preserve both linear dependence and linear independence.

Conjecture 1.1 asserts that the forms in (11) are linearly independent for every admissible marked configuration. For $n = 3$ the forms are binary quadratics. Let M be their coefficient matrix in the ordered basis

$$(Z_0^2, Z_0Z_1, Z_1^2).$$

Then $\beta_1, \beta_2, \beta_3$ are linearly dependent if and only if $\det M = 0$.

2.2. Retarded intersections with a complete timelike affine line. We next compute the retarded root determined by one observation event and one timelike affine emitter. For an observation event x and an emitter $\xi(s)$, put $q(s) = x - \xi(s)$. At the retarded point, $q(s)$ is future null.

Lemma 2.1 (Retarded point on a timelike affine line). *Let $\xi(s) = x_0 + sU$, where U is future-directed timelike, and put $\kappa^2 = -\langle U, U \rangle > 0$. If $x \notin \xi(\mathbb{R})$, set*

$$d = x - x_0, \quad A = \langle d, U \rangle, \quad R = \sqrt{A^2 + \kappa^2 \langle d, d \rangle}.$$

Then $R > 0$, and the past light cone of x meets ξ at exactly one point, namely $\xi(s^-)$ with

$$s^- = \frac{-A - R}{\kappa^2}. \quad (12)$$

The vector

$$q = x - \xi(s^-) = d + \frac{A + R}{\kappa^2} U \quad (13)$$

is future directed and null.

Proof. The null equation is

$$0 = \langle d - sU, d - sU \rangle = \langle d, d \rangle - 2As - \kappa^2 s^2,$$

whose roots are $(-A \pm R)/\kappa^2$. Choose a proper orthochronous Lorentz transformation sending U to $\kappa(1, \mathbf{0})$, and write the transformed displacement as $d = (\tau, \mathbf{r})$. Then $A = -\kappa\tau$, $R = \kappa|\mathbf{r}|$, and the two roots are

$$s^- = (\tau - |\mathbf{r}|)/\kappa, \quad s^+ = (\tau + |\mathbf{r}|)/\kappa.$$

At s^- one has $q = d - s^-\kappa(1, \mathbf{0}) = (|\mathbf{r}|, \mathbf{r})$, which is future null. At the other root, $\xi(s^+) - x = (|\mathbf{r}|, -\mathbf{r})$ is future null, so $\xi(s^+)$ lies on the future light cone of x . Since x does not lie on the emitter, $|\mathbf{r}| > 0$. The two algebraic roots therefore give one retarded and one advanced intersection. In particular, $R > 0$, and (12)–(13) follow. Transforming back preserves nullity and time orientation. $\square$

The proof of linear dependence needs only the retarded roots. The normalized determinant also requires reciprocal roots to be distinct; the following lemma supplies that fact for every affine configuration used here.

Lemma 2.2 (Reciprocal retarded rays are distinct). *Let ξ_i and ξ_j be disjoint complete timelike affine lines, with marked events $x_i \in \xi_i$ and $x_j \in \xi_j$. Then their two retarded directions satisfy $u_{ij} \neq u_{ji}$.*

Proof. Write

$$\xi_i(r) = x_i + rU_i, \quad \xi_j(s) = x_j + sU_j,$$

with U_i, U_j future timelike. Suppose, to the contrary, that the two retarded directions determine the same past-null ray $[w]_+$. For some $a, b > 0$ and some $r, s \in \mathbb{R}$, the two retarded emission equations would be

$$x_j + sU_j = x_i + aw, \quad x_i + rU_i = x_j + bw.$$

Adding them gives

$$rU_i + sU_j = (a + b)w. \tag{14}$$

If U_i and U_j are linearly independent, then $w \in \text{span}\{U_i, U_j\}$ by (14), and the first emission equation gives $x_j - x_i \in \text{span}\{U_i, U_j\}$. Writing $x_j - x_i = \alpha U_i + \beta U_j$ gives the common point $x_i + \alpha U_i = x_j - \beta U_j$, contrary to disjointness. If U_i and U_j are linearly dependent, then $U_j = \lambda U_i$ for some $\lambda > 0$, because both vectors are future directed. The left side of (14) is therefore $(r + \lambda s)U_i$, which is either zero or timelike, whereas the right side is nonzero and null. This is again impossible. $\square$

2.3. A common projective coordinate on the planar sky. We next pass from the retarded displacement to its incoming point on the common celestial sphere. For an ordered pair (i, j) , apply Lemma 2.1 with $x = x_i$ and $\xi = \xi_j$, and denote the resulting future null vector by q_{ij} . Atiyah's root is the opposite past-directed ray

$$u_{ij} = [\xi_j(s_{ij}^-) - x_i]_+ = [-q_{ij}]_+ \in \mathcal{S}^-.$$

Intersecting this ray with the section $t = -1$ gives the incoming sky direction

$$n_{ij} = -\frac{(q_{ij}^x, q_{ij}^y, q_{ij}^z)}{q_{ij}^0} \in S^2. \tag{15}$$

For a static emitter at spatial position $\mathbf{x}_j$, formula (15) gives $n_{ij} = (\mathbf{x}_j - \mathbf{x}_i)/|\mathbf{x}_j - \mathbf{x}_i|$, exactly the Euclidean direction in [3, Eq. (1.1)]. Thus (15) uses the same time orientation as Atiyah. As in Atiyah's definition, affine parallel translation identifies the null cones at the different events; no observer-dependent rest-frame aberration is applied.

The counterexample is planar, so $q_{ij}^z = 0$ and all six directions lie on the equator $S^1 \subset S^2$. We therefore choose a single real projective coordinate on that circle. This keeps the coefficient matrix

real and makes a sign-change argument possible. The following Cayley coordinate covers the entire equator, including its exceptional affine point:

$$C(n_x, n_y, 0) = \begin{cases} [1 + n_x : n_y], & (n_x, n_y) \neq (-1, 0), \\ [0 : 1], & (n_x, n_y) = (-1, 0). \end{cases} \quad (16)$$

Its inverse is the globally defined formula

$$C^{-1}([a : b]) = \left(\frac{a^2 - b^2}{a^2 + b^2}, \frac{2ab}{a^2 + b^2}, 0 \right). \quad (17)$$

Thus $C : S^1 \rightarrow \mathbb{P}^1(\mathbb{R})$ is a bijection; the second branch in (16) removes the apparent chart singularity at $(-1, 0, 0)$. To compare this real coordinate with the standard complex stereographic coordinate, we extend C once to the whole celestial sphere rather than choosing a coordinate separately for each observer. Let

$$\zeta_{\text{hom}}(n_x, n_y, n_z) = \begin{cases} [n_x + in_y : 1 - n_z], & (n_x, n_y, n_z) \neq (0, 0, 1), \\ [1 : 0], & (n_x, n_y, n_z) = (0, 0, 1), \end{cases}$$

and set

$$G_C = \begin{pmatrix} 1 & i \\ 1 & -i \end{pmatrix}, \quad \Phi = G_C^{-1} \circ \zeta_{\text{hom}}.$$

Here matrices act projectively on homogeneous column vectors. Lemma 2.3 verifies that $\Phi|_{S^1} = C$ and compares this global choice with the standard stereographic coordinate.

Whenever $q_{ij}^0 - q_{ij}^x > 0$, substituting (15) into (16) and multiplying both homogeneous coordinates by q^0 gives the convenient lift

$$C(n_{ij}) = \Phi(u_{ij}) = [h_{ij}^0 : h_{ij}^1], \quad h_{ij} = (q_{ij}^0 - q_{ij}^x, -q_{ij}^y). \quad (18)$$

Indeed, if $[h] = [a : b]$, then

$$(n_x, n_y) = \left(\frac{a^2 - b^2}{a^2 + b^2}, \frac{2ab}{a^2 + b^2} \right),$$

and, for $a = q^0 - q^x$ and $b = -q^y$, the null identity $(q^0)^2 = (q^x)^2 + (q^y)^2$ gives

$$\begin{aligned} a^2 + b^2 &= (q^0 - q^x)^2 + (q^y)^2 = 2q^0(q^0 - q^x), \\ a^2 - b^2 &= (q^0 - q^x)^2 - (q^y)^2 = -2q^x(q^0 - q^x), \\ 2ab &= -2q^y(q^0 - q^x). \end{aligned}$$

For a future planar null vector, $q^0 - q^x \geq 0$; equality forces $q^y = 0$ and corresponds to the exceptional branch of (16). On the complement of that point, division by the first identity yields

$$(n_x, n_y) = \left(-\frac{q^x}{q^0}, -\frac{q^y}{q^0} \right),$$

which is (15). Thus (18) gives all six roots in one projective coordinate. Section 4 verifies that these six lifts are nonzero. For a future planar null vector, nonvanishing is equivalent to $q_{ij}^0 - q_{ij}^x > 0$: equality forces $q_{ij}^y = 0$ and hence makes both coordinates in (18) vanish.

Lemma 2.3 (Comparison with standard stereographic projection). *Let $[a : b] \in \mathbb{P}^1(\mathbb{R})$ represent a planar sky direction by (18). Use stereographic projection from the north pole $(0, 0, 1)$, with affine coordinate $\zeta = (n_x + in_y)/(1 - n_z)$. On the equator this becomes*

$$\zeta = n_x + in_y = \frac{a + ib}{a - ib}. \quad (19)$$

Consequently, the passage from our six roots h_{ij} to their standard stereographic coordinates is induced by the single matrix $G_C \in \text{GL}(2, \mathbb{C})$ displayed above:

$$[a : b] \longmapsto [a + ib : a - ib].$$

In particular, the three homogeneous quadratics are linearly dependent in our real coordinate if and only if they are linearly dependent in the standard stereographic coordinate.

Proof. The displayed formula for (n_x, n_y) gives

$$n_x + in_y = \frac{a^2 - b^2 + 2iab}{a^2 + b^2} = \frac{a + ib}{a - ib},$$

which proves (19). The determinant of G_C is $-2i$. The corresponding contragredient change of variables therefore induces an invertible linear map on the three-dimensional vector space of binary quadratics. Applying the same map to all three polynomials preserves both linear dependence and linear independence. $\square$

We have now fixed the causal and projective input once and for all. For the explicit family, every ordered pair will therefore produce a unique root in one common projective coordinate, and dependence may be tested by an ordinary real coefficient determinant.

3. THE EXPLICIT THREE-WORLDFAMILY

We now verify the causal properties of the family (1)–(3). Only the temporal placement of the third line varies; the velocities and spatial tracks remain fixed. The parameter therefore isolates the effect of retarded timing from collisions and changes of velocity. For $T \in I$, let ξ_i be (1)–(3) with T in place of T_* , and let $x_i = \xi_i(0)$. The three marked events are distinct because their spatial coordinates are distinct. The direction vectors are

$$U_1 = (1, 0, 0), \quad U_2 = \left(1, \frac{1}{2}, -\frac{1}{10}\right), \quad U_3 = \left(1, \frac{4}{5}, -\frac{1}{5}\right). \quad (20)$$

These vectors are future directed and

$$\langle U_1, U_1 \rangle = -1, \quad \langle U_2, U_2 \rangle = -\frac{37}{50}, \quad \langle U_3, U_3 \rangle = -\frac{8}{25}. \quad (21)$$

The spatial speeds are 0, $\sqrt{26}/10$, and $\sqrt{17}/5$; each is strictly less than one. Reparametrizing each line by proper time changes neither the underlying worldline nor any celestial direction.

Proposition 3.1 (Causal admissibility of the family). *For every $T \in I$, the three complete affine lines ξ_1, ξ_2, ξ_3 are pairwise disjoint. Hence each of the six ordered pairs has a unique retarded intersection.*

Proof. An intersection of ξ_1 and ξ_2 would require both spatial coordinates of $\xi_2(r)$ to vanish. The first forces $r = 1/2$, whereas the second then equals $3/20$. An intersection of ξ_1 and ξ_3 would force $r = -5/4$ from the first spatial coordinate, whereas the second would then equal $3/8$.

For $\xi_2(r)$ and $\xi_3(s)$, equality of the two spatial coordinates has the equations

$$-\frac{1}{4} + \frac{r}{2} = 1 + \frac{4s}{5}, \quad \frac{1}{5} - \frac{r}{10} = \frac{1}{8} - \frac{s}{5},$$

or, after clearing denominators,

$$10r - 16s = 25, \quad 4r - 8s = 3.$$

The second equation gives $r = 2s + 3/4$; substitution in the first gives

$$4s + \frac{15}{2} = 25, \quad s = \frac{35}{8}, \quad r = \frac{19}{2}.$$

At this unique candidate pair of parameters, the two time coordinates are

$$-\frac{2}{3} + \frac{19}{2} = \frac{53}{6}, \quad T + \frac{35}{8}.$$

Equality of the time coordinates would force

$$T = \frac{53}{6} - \frac{35}{8} = \frac{107}{24} > \frac{9}{10}.$$

Thus the lines are pairwise disjoint, and Lemma 2.1 supplies the six unique retarded points. $\square$

4. FROM SIX RETARDED ROOTS TO ONE DETERMINANT

This section performs the algebraic reduction that drives the proof. We first express the coefficient determinant directly in terms of the six projective roots, then compute those roots from the retarded-intersection formula, and finally verify that the resulting determinant is a smooth real function throughout the parameter interval.

4.1. The six-bracket identity. We begin by reducing linear dependence to one explicit determinant. For a nonzero vector $a = (a_0, a_1) \in \mathbb{C}^2$, write

$$L_a(Z_0, Z_1) = a_0 Z_1 - a_1 Z_0,$$

and, for nonzero $a, b \in \mathbb{C}^2$, set $p_{a,b} = L_a L_b$. Since $L_{h_{ij}} = -\ell_{ij}$, the two signs cancel and, for $n = 3$, $p_{h_{ij}, h_{ik}}$ is exactly β_i . Its coefficient row in the ordered basis $(Z_0^2, Z_0 Z_1, Z_1^2)$ is

$$r(a, b) = (a_1 b_1, -a_0 b_1 - a_1 b_0, a_0 b_0). \quad (22)$$

Write $[a, b] = a_0 b_1 - a_1 b_0$.

Lemma 4.1 (Bracket identity). *For $a, b, c, d, e, f \in \mathbb{C}^2$,*

$$\det \begin{pmatrix} r(a, b) \\ r(c, d) \\ r(e, f) \end{pmatrix} = [a, d][b, f][c, e] + [a, e][b, c][d, f]. \quad (23)$$

Proof. Both sides are polynomial in the six vectors. Under a common $G \in \text{GL}(2, \mathbb{C})$, the right-hand side is multiplied by $\det(G)^3$, one factor for each bracket. For the left-hand side, write $Z = (Z_0, Z_1)$ and observe that

$$L_{Ga}(Z) = \det(G) L_a(G^{-1}Z)$$

gives

$$p_{Ga, Gb}(Z) = \det(G)^2 p_{a,b}(G^{-1}Z).$$

Thus each coefficient row acquires the scalar $\det(G)^2$, while all three rows undergo the same change of variables $\text{Sym}^2(G^{-1})$. Their determinant is therefore multiplied by

$$\det(G)^6 \det \text{Sym}^2(G^{-1}) = \det(G)^6 \det(G)^{-3} = \det(G)^3.$$

On the dense set $[a, b] \neq 0$, choose $G \in \text{GL}(2, \mathbb{C})$ sending a and b to the standard basis. Because both sides acquire the same nonzero factor, it is enough to verify the identity after this transformation. For $a = (1, 0)$ and $b = (0, 1)$ one has $r(a, b) = (0, -1, 0)$, and the determinant on the left becomes

$$c_1 d_1 e_0 f_0 - c_0 d_0 e_1 f_1.$$

The right-hand side gives the same expression. Since both sides are polynomial in the six vectors, equality on this dense set proves the identity everywhere. $\square$

Apply the lemma with $(a, b, c, d, e, f) = (h_{12}, h_{13}, h_{21}, h_{23}, h_{31}, h_{32})$. The three rows are the coefficient rows of $\beta_1, \beta_2, \beta_3$, so these quadratics are dependent exactly when

$$\Delta_h = [h_{12}, h_{23}][h_{13}, h_{32}][h_{21}, h_{31}] + [h_{12}, h_{31}][h_{13}, h_{21}][h_{23}, h_{32}] \quad (24)$$

vanishes. Rescaling the six lifts multiplies Δ_h by the product of the six scale factors. In particular, positive rescaling preserves both its zero set and its sign.

Equation (24) is the central algebraic bridge in the paper: it reduces Theorem 1.2 to the concrete task of computing the six roots in the common Cayley coordinate and showing that the two triple products cancel for some $T \in I$.

4.2. Retarded parameters and exact root lifts. All square roots below denote positive real roots. To keep the radicands readable, set

$$P_1(T) = 1088T^2 - 2480T + 1481, \quad (25)$$

$$P_2(T) = 244800T^2 - 404400T + 173129, \quad (26)$$

$$P_3(T) = 14976T^2 - 52896T + 47963. \quad (27)$$

Lemma 2.1 uses the displacement from the emitter's base point to the observation event. In the present family these six vectors are

$$\begin{aligned} d_{12} &= \left(\frac{2}{3}, \frac{1}{4}, -\frac{1}{5}\right), & d_{13} &= \left(-T, -1, -\frac{1}{8}\right), \\ d_{21} &= \left(-\frac{2}{3}, -\frac{1}{4}, \frac{1}{5}\right), & d_{23} &= \left(-T - \frac{2}{3}, -\frac{5}{4}, \frac{3}{40}\right), \\ d_{31} &= \left(T, 1, \frac{1}{8}\right), & d_{32} &= \left(T + \frac{2}{3}, \frac{5}{4}, -\frac{3}{40}\right). \end{aligned}$$

Accordingly, let $A_{ij} = \langle d_{ij}, U_j \rangle$, $\kappa_j^2 = -\langle U_j, U_j \rangle$, and let R_{ij} denote the positive number R from Lemma 2.1, namely

$$R_{ij} := \sqrt{A_{ij}^2 + \kappa_j^2 \langle d_{ij}, d_{ij} \rangle}.$$

Substitution into the retarded formula gives the six scalar products

$$\begin{aligned} A_{12} &= -\frac{2}{3} + \frac{1}{8} + \frac{1}{50} = -\frac{313}{600}, \\ A_{13} &= T - \frac{4}{5} + \frac{1}{40} = \frac{40T - 31}{40}, \\ A_{21} &= \frac{2}{3}, \\ A_{23} &= T + \frac{2}{3} - 1 - \frac{3}{200} = \frac{600T - 209}{600}, \\ A_{31} &= -T, \\ A_{32} &= -T - \frac{2}{3} + \frac{5}{8} + \frac{3}{400} = -\frac{1200T + 41}{1200}. \end{aligned}$$

The corresponding radicands are

$$\begin{aligned}
R_{12}^2 &= \left(-\frac{313}{600}\right)^2 + \frac{37}{50} \left(-\frac{4}{9} + \frac{1}{16} + \frac{1}{25}\right) = \frac{11}{576}, \\
R_{13}^2 &= \left(T - \frac{31}{40}\right)^2 + \frac{8}{25} \left(-T^2 + 1 + \frac{1}{64}\right) = \frac{P_1(T)}{1600}, \\
R_{21}^2 &= \left(\frac{2}{3}\right)^2 - \frac{4}{9} + \frac{1}{16} + \frac{1}{25} = \frac{41}{400}, \\
R_{23}^2 &= \left(T - \frac{209}{600}\right)^2 + \frac{8}{25} \left[-\left(T + \frac{2}{3}\right)^2 + \frac{25}{16} + \frac{9}{1600}\right] = \frac{P_2(T)}{360000}, \\
R_{31}^2 &= T^2 - T^2 + 1 + \frac{1}{64} = \frac{65}{64}, \\
R_{32}^2 &= \left(T + \frac{41}{1200}\right)^2 + \frac{37}{50} \left[-\left(T + \frac{2}{3}\right)^2 + \frac{25}{16} + \frac{9}{1600}\right] = \frac{P_3(T)}{57600}.
\end{aligned}$$

These scalar quantities determine the retarded parameters, but the coefficient determinant requires actual projective lifts. Put $\lambda_{ij} = (A_{ij} + R_{ij})/\kappa_j^2$. The preceding identities give

$$\begin{aligned}
\lambda_{12} &= \frac{-313 + 25\sqrt{11}}{444}, & \lambda_{13} &= \frac{5(40T - 31 + \sqrt{P_1})}{64}, \\
\lambda_{21} &= \frac{2}{3} + \frac{\sqrt{41}}{20}, & \lambda_{23} &= \frac{600T - 209 + \sqrt{P_2}}{192}, \\
\lambda_{31} &= -T + \frac{\sqrt{65}}{8}, & \lambda_{32} &= \frac{-1200T - 41 + 5\sqrt{P_3}}{888}.
\end{aligned}$$

Equations (13) and (18) then give one formula for all six projective lifts:

$$h_{ij} = \left(d_{ij}^0 - d_{ij}^x + \frac{A_{ij} + R_{ij}}{\kappa_j^2}(U_j^0 - U_j^x), -d_{ij}^y - \frac{A_{ij} + R_{ij}}{\kappa_j^2}U_j^y\right). \quad (28)$$

For example, consider the light seen by observer 1 from worldline 2. The quantities above read

$$d_{12} = \left(\frac{2}{3}, \frac{1}{4}, -\frac{1}{5}\right), \quad \kappa_2^2 = \frac{37}{50}, \quad A_{12} = -\frac{313}{600}, \quad R_{12} = \frac{\sqrt{11}}{24}.$$

Hence

$$\lambda_{12} = \frac{A_{12} + R_{12}}{\kappa_2^2} = \frac{-313 + 25\sqrt{11}}{444}.$$

Since $U_2 = (1, 1/2, -1/10)$, formula (28) gives

$$h_{12} = \left(\frac{5}{12} + \frac{\lambda_{12}}{2}, \frac{1}{5} + \frac{\lambda_{12}}{10}\right) = \frac{1}{888}(57 + 25\sqrt{11}, 5(\sqrt{11} + 23)).$$

Thus the six lifts are obtained uniformly through the chain

$$d \mapsto (A, R) \mapsto \lambda \mapsto q \mapsto h.$$

Applying the same formula to the other ordered pairs gives the complete list below. In this display, and until the end of the section, we suppress the argument T in $P_1(T), P_2(T), P_3(T)$. For each lift, the first line records the direct substitution and the second its simplified form.

$$\begin{aligned}
h_{12} &= \left(\frac{5}{12} + \frac{\lambda_{12}}{2}, \frac{1}{5} + \frac{\lambda_{12}}{10} \right) \\
&= \frac{1}{888} (57 + 25\sqrt{11}, 5(\sqrt{11} + 23)),
\end{aligned} \tag{29}$$

$$\begin{aligned}
h_{13} &= \left(1 - T + \frac{\lambda_{13}}{5}, \frac{1}{8} + \frac{\lambda_{13}}{5} \right) \\
&= \frac{1}{64} (\sqrt{P_1} + 33 - 24T, \sqrt{P_1} + 40T - 23),
\end{aligned} \tag{30}$$

$$\begin{aligned}
h_{21} &= \left(-\frac{5}{12} + \lambda_{21}, -\frac{1}{5} \right) \\
&= \frac{1}{20} (5 + \sqrt{41}, -4),
\end{aligned} \tag{31}$$

$$\begin{aligned}
h_{23} &= \left(-T + \frac{7}{12} + \frac{\lambda_{23}}{5}, -\frac{3}{40} + \frac{\lambda_{23}}{5} \right) \\
&= \frac{1}{960} (\sqrt{P_2} + 351 - 360T, \sqrt{P_2} + 600T - 281),
\end{aligned} \tag{32}$$

$$\begin{aligned}
h_{31} &= \left(T - 1 + \lambda_{31}, -\frac{1}{8} \right) \\
&= \frac{1}{8} (\sqrt{65} - 8, -1),
\end{aligned} \tag{33}$$

$$\begin{aligned}
h_{32} &= \left(T - \frac{7}{12} + \frac{\lambda_{32}}{2}, \frac{3}{40} + \frac{\lambda_{32}}{10} \right) \\
&= \frac{1}{1776} (576T + 5\sqrt{P_3} - 1077, \\
&\quad \sqrt{P_3} + 125 - 240T).
\end{aligned} \tag{34}$$

Define the numerator lifts H_{ij} by clearing the six positive denominators in (29)–(34). Thus

$$(h_{12}, h_{13}, h_{21}, h_{23}, h_{31}, h_{32}) = \left(\frac{H_{12}}{888}, \frac{H_{13}}{64}, \frac{H_{21}}{20}, \frac{H_{23}}{960}, \frac{H_{31}}{8}, \frac{H_{32}}{1776} \right). \tag{35}$$

4.3. A smooth real determinant. For the interval argument, we must verify that these formulas give continuous, nonzero lifts throughout I . We first check that every radicand is positive. The derivative bounds

$$\begin{aligned}
P_1'(T) &= 2176T - 2480 \leq P_1'(9/10) = -\frac{2608}{5} < 0, \\
P_3'(T) &= 29952T - 52896 \leq P_3'(9/10) = -\frac{129696}{5} < 0
\end{aligned}$$

show that P_1 and P_3 are decreasing on I . Moreover, $P_2''(T) = 489600 > 0$ and

$$P_2'(T) = 0 \iff T = \frac{337}{408} \in I.$$

Their minima on I are therefore

$$P_1(9/10) = \frac{3257}{25}, \quad P_2(337/408) = \frac{103968}{17}, \quad P_3(9/10) = \frac{312179}{25}. \tag{36}$$

It remains to check that none of the six lifts vanishes. For H_{13} and H_{23} , the first coordinates satisfy

$$H_{13}^0 \geq \sqrt{P_1(T)} + 33 - 24(9/10) > \frac{57}{5},$$

$$H_{23}^0 \geq \sqrt{P_2(T)} + 351 - 360(9/10) > 27.$$

The second coordinate of H_{13} also satisfies

$$H_{13}^1 = \sqrt{P_1(T)} + 40T - 23 > 40(3/4) - 23 = 7.$$

For H_{32} , the inequality $111^2 < 312179/25$ gives

$$H_{32}^1 \geq \sqrt{312179/25} + 125 - 240(9/10) > 20.$$

Finally, both coordinates of H_{12} are positive, while $H_{21}^1 = -4$ and $H_{31}^1 = -1$. Hence none of the six vectors vanishes.

We now compare the coefficient matrices associated with the two choices of lifts. Let $M_h(T)$ be the matrix whose rows are

$$r(h_{12}, h_{13}), \quad r(h_{21}, h_{23}), \quad r(h_{31}, h_{32}),$$

and define $M_H(T)$ analogously from the numerator lifts H_{ij} . Put $\Delta_h = \det M_h$ and $\Delta_H = \det M_H$. These are smooth real functions on I . Multilinearity and (35) give

$$\Delta_H = (888)(64)(20)(960)(8)(1776) \Delta_h.$$

The factor is positive and constant. Thus the two determinants have the same zeros and the same sign, and a zero is simple for one if and only if it is simple for the other.

5. A UNIQUE SIMPLE DETERMINANT CROSSING

The next proposition concerns the determinant $\Delta_H = \det M_H$ and supplies the determinant zero, monotonicity, and root separation needed to identify that zero as a nondegenerate rank-two crossing. It is proved in Appendix A by exact rational interval estimates.

Proposition 5.1 (Exact crossing and separation of the six roots). *With T_-, T_+, J as in Section 1, one has*

$$\Delta_H(T_-) > 0, \quad \Delta_H(T_+) < 0, \quad \Delta'_H(T) < 0 \quad (T \in J).$$

Moreover, every bracket between two distinct vectors among

$$H_{12}(T), H_{13}(T), H_{21}(T), H_{23}(T), H_{31}(T), H_{32}(T)$$

is nonzero for every $T \in J$.

Proof of Theorem 1.2. The strict endpoint signs in Proposition 5.1 and the intermediate value theorem give a zero $T_* \in (T_-, T_+)$. Since $\Delta'_H < 0$ throughout J , the function is strictly decreasing there; the zero is unique in J , and $\Delta'_H(T_*) \neq 0$. The positive constant relating Δ_H and Δ_h gives the same conclusions for Δ_h .

The simplicity statement is independent of the chosen smooth nonzero lift normalization. Indeed, independently rescaling the six lifts by smooth nonvanishing functions multiplies the determinant by their product, so the new determinant has the form $\tilde{\Delta}(T) = g(T)\Delta_h(T)$ with $g(T) \neq 0$. At a zero of Δ_h one therefore has

$$\tilde{\Delta}'(T_*) = g(T_*)\Delta'_h(T_*).$$

A fixed common projective coordinate change contributes only an additional nonzero constant factor to the coefficient determinant and likewise preserves simplicity.

Proposition 3.1 and (21) show that all three worldlines are complete, pairwise disjoint, future directed, and timelike throughout $[3/4, 9/10]$; the spatial speeds computed after (21) are strictly subluminal. Lemma 2.1 gives the six unique retarded intersections. By the final assertion of

Proposition 5.1, the six projective roots at T_* are pairwise distinct. In particular, the three two-element root multisets are disjoint. Proportional nonzero homogeneous quadratics have the same root multiset, with multiplicities; hence no two rows of the coefficient matrix are proportional. The matrix is singular but has rank at least two, and therefore has rank exactly two.

At a rank-two matrix M , the adjugate $\text{adj}(M)$ is nonzero, so $d(\det)_M(E) = \text{tr}(\text{adj}(M)E)$ is a nonzero linear functional. Thus the determinant-zero set in $M_3(\mathbb{R})$ is a smooth hypersurface near $M_H(T_*)$. Since

$$\left. \frac{d}{dT} \det M_H(T) \right|_{T=T_*} = \Delta'_H(T_*) \neq 0,$$

the path $T \mapsto M_H(T)$ meets that hypersurface transversely. Finally, the common $\text{GL}(2, \mathbb{C})$ transformation in Lemma 2.3 preserves linear dependence and identifies the same counterexample in the standard stereographic coordinate. $\square$

6. NULL-TRANSLATION STABILIZATION AT EVERY PARTICLE NUMBER

The three-line crossing is the essential counterexample. To obtain a separate counterexample at each prescribed particle number, we now build a geometric mechanism that preserves the three-term relation while adding worldlines. The argument has three parts: an algebraic common-factor lemma, a general Lorentzian realization, and an application of the general criterion to the explicit three-line core.

6.1. The common-factor mechanism. To distinguish the three-worldline forms from those of an enlarged configuration, write

$$\beta_i^{(m)} = \prod_{\substack{1 \leq j \leq m \\ j \neq i}} \ell_{ij} \in \text{Sym}^{m-1}((\mathbb{C}^2)^*). \quad (37)$$

Lemma 6.1 (Common-factor preservation). *Let $N > 3$. Suppose an admissible three-worldline configuration satisfies the nontrivial relation*

$$\lambda_1 \beta_1^{(3)} + \lambda_2 \beta_2^{(3)} + \lambda_3 \beta_3^{(3)} = 0. \quad (38)$$

If these three worldlines are contained in an admissible N -worldline configuration and, for one celestial direction u_ ,*

$$u_{ik} = u_*, \quad i \in \{1, 2, 3\}, \quad k \in \{4, \dots, N\},$$

then the N forms are linearly dependent.

Proof. Choose one nonzero linear factor L_* vanishing at $\Phi(u_*)$. Retain the six old lifts and use L_* for every occurrence of the common new root. With these choices, the identities

$$\beta_i^{(N)} = L_*^{N-3} \beta_i^{(3)}, \quad i = 1, 2, 3,$$

hold exactly. Multiplication of (38) by L_*^{N-3} gives a nontrivial relation among the N forms, with zero coefficients on the new ones. For arbitrary lift choices, each form is merely rescaled by a nonzero number, and the coefficients of the relation may be rescaled inversely. Thus the conclusion is intrinsic. $\square$

6.2. A general null-translation realization. The common-factor argument has a geometric realization that is independent of the particular numerical example. Translating all worldlines and marked events by the same vector leaves every retarded displacement, and hence every celestial root, unchanged. We may therefore formulate the criterion using a linear two-plane through the origin.

Proposition 6.2 (Null-translation stabilization principle). *Let $N > 3$, and let $\xi_i(r) = x_i + rU_i$, $i = 1, 2, 3$, be pairwise disjoint complete timelike affine lines, with each U_i future directed, whose associated forms are dependent. Suppose that the marked events lie in the two-plane*

$$\Pi = \text{span}_{\mathbb{R}}\{U_*, \omega\},$$

where U_* is future-directed timelike and ω is a nonzero past-directed null vector, and suppose $U_i \notin \Pi$ for $i = 1, 2, 3$. Write

$$x_i = \alpha_i U_* + \rho_i \omega.$$

For any distinct real numbers $c_4, \dots, c_N$ satisfying $c_k > \max_i \rho_i$, the additional lines

$$\xi_k(r) = rU_* + c_k \omega, \quad x_k = c_k \omega, \quad 4 \leq k \leq N, \quad (39)$$

can be adjoined to the original three lines to produce an admissible N -worldline configuration with linearly dependent forms.

Proof. The vectors U_* and ω are independent, because a nonzero null vector cannot be proportional to a timelike vector. Hence the new parallel lines are mutually disjoint. Each new line lies in Π . Since $x_i \in \Pi$ and $U_i \notin \Pi$, the old line $x_i + \mathbb{R}U_i$ meets Π only at x_i . In the basis (U_*, ω) , the point x_i has ω -coordinate ρ_i , whereas every point of the k th new line has ω -coordinate c_k . Since $c_k > \rho_i$, the point x_i does not lie on the k th new line. Any intersection of an old line with a new one would lie in Π and hence would have to be the marked point x_i ; the coordinate comparison just made rules this out. Thus no old–new intersection occurs. Together with the pairwise disjointness of the old lines and of the new lines, this proves that all N lines are pairwise disjoint. They are complete and timelike, so Lemma 2.1 gives admissibility.

For $i \leq 3 < k$, the point

$$y_{ik} = \alpha_i U_* + c_k \omega \in \xi_k$$

satisfies

$$y_{ik} - x_i = (c_k - \rho_i)\omega.$$

Because $c_k - \rho_i > 0$, this is a nonzero past-null displacement. Uniqueness in Lemma 2.1 makes it the retarded displacement, and consequently $u_{ik} = [\omega]_+$ for every $i \leq 3 < k$. Adjoining the new lines does not alter any of the old worldlines, marked events, or old–old retarded intersections. Hence the six old factors and the original three-term relation are unchanged. The conclusion then follows from Lemma 6.1. $\square$

6.3. Application to the three-line construction. We now verify the hypotheses for the three-line core. Fix the parameter T_* from Theorem 1.2. Its marked events are

$$x_1 = (0, 0, 0), \quad x_2 = \left(-\frac{2}{3}, -\frac{1}{4}, \frac{1}{5}\right), \quad x_3 = \left(T_*, 1, \frac{1}{8}\right). \quad (40)$$

Set

$$U_* = -x_2 = \left(\frac{2}{3}, \frac{1}{4}, -\frac{1}{5}\right), \quad \kappa_*^2 = -\langle U_*, U_* \rangle = \frac{1231}{3600}. \quad (41)$$

Thus U_* is future directed and timelike. We next choose a null vector in the marked-event plane. Specifically, we solve $\langle x_3 - \sigma U_*, x_3 - \sigma U_* \rangle = 0$. Set

$$A_* = \langle x_3, U_* \rangle = \frac{27 - 80T_*}{120}, \quad B_* = \langle x_3, x_3 \rangle = \frac{65}{64} - T_*^2, \quad (42)$$

$$\Theta_* = 23616T_*^2 - 69120T_* + 91679, \quad R_* = \sqrt{A_*^2 + \kappa_*^2 B_*} = \frac{\sqrt{\Theta_*}}{480}, \quad (43)$$

$$\sigma_* = \frac{R_* - A_*}{\kappa_*^2}, \quad \omega = x_3 - \sigma_* U_*. \quad (44)$$

Because $B_* > 65/64 - 81/100 > 0$, one has $R_* > |A_*|$ and $\sigma_* > 0$.

Lemma 6.3 (The stabilizing plane). *The vector ω is nonzero, null, and past directed, and*

$$\Pi := \text{span}_{\mathbb{R}}\{x_2, x_3\} = \text{span}_{\mathbb{R}}\{U_*, \omega\}.$$

Moreover, none of the three original direction vectors lies in Π .

Proof. The definition of σ_* and the identity $R_*^2 = A_*^2 + \kappa_*^2 B_*$ give

$$\langle \omega, \omega \rangle = B_* - 2A_*\sigma_* - \kappa_*^2\sigma_*^2 = 0.$$

Also

$$\langle U_*, \omega \rangle = A_* + \kappa_*^2\sigma_* = R_* > 0.$$

In the rest frame of the future timelike vector U_* , this inequality says that the time component of ω is negative. Hence ω is past directed, and in particular nonzero. Since a nonzero null vector cannot be proportional to a timelike vector, the identities $x_2 = -U_*$ and $x_3 = \sigma_* U_* + \omega$ prove the equality of the two spanning planes.

For the original directions U_i in (20), direct calculation gives

$$\begin{aligned} \det[x_2 \ x_3 \ U_1] &= -\frac{37}{160}, \\ \det[x_2 \ x_3 \ U_2] &= \frac{36T_* - 59}{480} < 0, \\ \det[x_2 \ x_3 \ U_3] &= \frac{88T_* - 25}{800} > 0. \end{aligned}$$

The signs use $3/4 < T_* < 9/10$. Thus $U_i \notin \Pi$ for all i . $\square$

Proof of Theorem 1.3. The case $N = 3$ is Theorem 1.2. Let $N > 3$. In the basis (U_*, ω) , the three marked events are

$$x_1 = 0, \quad x_2 = -U_*, \quad x_3 = \sigma_* U_* + \omega,$$

so $(\rho_1, \rho_2, \rho_3) = (0, 0, 1)$. Choose $c_k = k - 2$ for $k = 4, \dots, N$. These numbers are distinct and satisfy $c_k > \max_i \rho_i = 1$. Proposition 6.2 therefore produces the required admissible N -worldline configuration and preserves the three-term dependence. $\square$

Remark 6.4 (Genericity of the stabilized examples). For $N > 3$, the first three forms acquire the repeated common factor L_*^{N-3} . The worldlines remain pairwise disjoint; coincident celestial directions are permitted by Atiyah's conjecture. A stronger variant requiring all roots to be distinct is a different problem. The three-worldline core itself has six pairwise distinct roots by Proposition 5.1.

7. GEOMETRY OF THE RANK DEFECT, NORMALIZATION, AND DISCUSSION

7.1. Projective balance and Veronese concurrence. For six nonzero lifts, the bracket identity (24) has a projective reformulation. Whenever its second triple product is nonzero, set

$$\mathcal{R} = \frac{[h_{12}, h_{23}][h_{13}, h_{32}][h_{21}, h_{31}]}{[h_{12}, h_{31}][h_{13}, h_{21}][h_{23}, h_{32}]}.$$
 (45)

Proposition 7.1 (Projective balance at the crossing). *The quantity $\mathcal{R}$ is invariant under independent rescaling of the six lifts and under a simultaneous element of $\text{PGL}(2, \mathbb{C})$. At the parameter T_* of Theorem 1.2,*

$$\mathcal{R}(T_*) = -1.$$

The space of linear relations among $\beta_1, \beta_2, \beta_3$ is one-dimensional, and every nonzero relation $c_1\beta_1 + c_2\beta_2 + c_3\beta_3 = 0$ satisfies $c_1c_2c_3 \neq 0$.

Proof. Each lift occurs once in the numerator and once in the denominator of (45); a common $G \in \mathrm{GL}(2, \mathbb{C})$ multiplies every bracket by $\det G$. Thus $\mathcal{R}$ is projectively invariant. Proposition 5.1 makes all six roots pairwise distinct, so every bracket in (45) is nonzero. Dividing (24) by the second triple product gives $\Delta_h = 0$ if and only if $\mathcal{R} = -1$.

The coefficient matrix has rank two by Theorem 1.2, so its relation space is one-dimensional. If one coefficient in a nonzero relation vanished, the remaining two quadratics would be proportional, contrary to the fact that their root multisets are pairwise disjoint. $\square$

The same rank defect has a classical projective interpretation. Use the natural pairing of $\mathrm{Sym}^2((\mathbb{R}^2)^*)$ with $\mathrm{Sym}^2(\mathbb{R}^2)$. A binary quadratic p then defines the line

$$L_p = \{[w] \in \mathbb{P}(\mathrm{Sym}^2(\mathbb{R}^2)) : \langle p, w \rangle = 0\}.$$

Consider the Veronese map

$$\nu_2 : \mathbb{P}(\mathbb{R}^2) \longrightarrow \mathbb{P}(\mathrm{Sym}^2(\mathbb{R}^2)), \quad [v] \longmapsto [v \odot v],$$

where $\odot$ denotes the symmetric product. If p has two distinct roots $[a]$ and $[b]$, then the line L_p is precisely the secant through $\nu_2([a])$ and $\nu_2([b])$. The rank-two conclusion therefore says that the three secants associated with the three observers are concurrent.

At the same parameter, the coefficient-matrix path $T \mapsto M_H(T)$ meets the smooth rank-two stratum of $\{\det = 0\}$ in $M_3(\mathbb{R})$. Since $\Delta'_H(T_*) \neq 0$, the intersection is transverse. We make no transversality claim in the full complex ambient space, where the determinantal divisor has real codimension two. Thus the failure is not caused by a collision, a lightlike limit, a repeated root, or a higher-order tangency along the displayed real path. Within this family, it is a rank defect produced solely by retarded timing inside the strictly timelike inertial region.

7.2. The normalized determinant and the affine-timelike dichotomy. For an admissible N -tuple whose reciprocal roots are distinct, choose lifts h_{ij} and let C_N be the matrix whose i th row is the coefficient row of $\beta_i^{(N)}$ in the basis $(Z_0^{N-1}, Z_0^{N-2}Z_1, \dots, Z_1^{N-1})$. Define

$$D_{\mathrm{ret}}^{(N)} = \frac{\det C_N}{\prod_{1 \leq i < j \leq N} [h_{ij}, h_{ji}]}. \quad (46)$$

The quotient in (46) is the retarded form of the normalized determinant developed by Atiyah and Sutcliffe. Their construction divides the coefficient determinant by the product of reciprocal-pair brackets, and they apply the same normalized function to Minkowski retarded data [5, §§3,10]. Our displayed basis and bracket conventions fix one representative of this normalization; changing those global conventions can multiply the quotient by a fixed nonzero scalar, but cannot change its zero set.

Corollary 7.2 (Normalized zeros for every $N \geq 3$). *For every $N \geq 3$ and every configuration constructed in the proof of Theorem 1.3, the quotient (46) is well defined, independent of the chosen lifts, invariant under a common $\mathrm{PGL}(2, \mathbb{C})$ coordinate change, and equal to zero.*

Proof. Lemma 2.2 makes every denominator factor in (46) nonzero. Each ordered lift h_{ij} occurs exactly once in the product defining $\beta_i^{(N)}$ and exactly once in the reciprocal bracket $[h_{ij}, h_{ji}]$. Rescaling that lift therefore multiplies the corresponding polynomial row and its unique denominator bracket by the same scalar, so the quotient is independent of the chosen lifts.

Under a common $G \in \mathrm{GL}(2, \mathbb{C})$, each degree- $(N-1)$ form acquires the scalar $\det(G)^{N-1}$ and undergoes the common substitution $\mathrm{Sym}^{N-1}(G^{-1})$. Factoring the scalar from all N rows contributes $\det(G)^{N(N-1)}$ to the coefficient determinant. Moreover,

$$\det \mathrm{Sym}^{N-1}(G^{-1}) = \det(G)^{-N(N-1)/2};$$

for diagonalizable G this follows by multiplying the N symmetric-power weights, and the general case follows by density. The numerator therefore acquires the net factor $\det(G)^{N(N-1)/2}$. The denominator contains $N(N-1)/2$ brackets and acquires the same factor, so the quotient is projectively invariant. An anti-Möbius change is a projective linear change followed by simultaneous conjugation; it therefore conjugates the quotient and, in particular, preserves its vanishing. Finally, Theorem 1.3 makes the rows of C_N linearly dependent, so $\det C_N = 0$. $\square$

Corollary 7.3 (Dichotomy for complete timelike affine lines). *Within the class of pairwise disjoint complete future-directed timelike affine lines, Atiyah's universal independence assertion holds for $n = 2$ and fails for every $n \geq 3$.*

Proof. The failure for every $n \geq 3$ is Theorem 1.3. For $n = 2$, the two forms are linear and vanish at u_{12} and u_{21} , respectively. Those roots are distinct by Lemma 2.2; hence the two forms are not proportional and are therefore linearly independent. $\square$

7.3. What the counterexamples do and do not settle. The counterexamples refute Atiyah's printed universal assertion and its uniform-motion interpretation [3, §1.6]. The three-worldline core is nondegenerate in the following concrete sense: its six ordered roots are pairwise distinct, its coefficient matrix has rank exactly two, and the displayed real path crosses the smooth rank-two stratum of the determinant hypersurface transversely. For $N > 3$, the stabilization intentionally introduces repeated common roots, which are permitted by Atiyah's formulation [3, Conjecture 1.6.1]. Whether counterexamples exist for every N with all ordered roots pairwise distinct is a stronger genericity problem and is not settled here.

No simultaneous-observation condition appears in Atiyah's printed Conjecture 1.6.1 [3]. The marked events in our construction cannot be simultaneous in any inertial frame: one has

$$\langle x_2 - x_1, x_2 - x_1 \rangle = -\frac{1231}{3600} < 0,$$

whereas the difference of two distinct points on a spacelike affine hyperplane is spacelike. Requiring synchronized observations would therefore define a different, stronger problem. Finally, the examples are neither static nor formed by inertial rays from a common event. Atiyah identified those two specializations with the Euclidean and hyperbolic branches, respectively; see [2, §6] and [3, §1.6]. Consequently, the present counterexamples do not address either of those separate conjectures.

APPENDIX A. PROOF OF PROPOSITION 5.1

This appendix proves Proposition 5.1. Every terminating decimal below is interpreted as the corresponding rational number. For intervals $X = [x_-, x_+]$ and $Y = [y_-, y_+]$, we use the standard rational operations

$$\begin{aligned} X + Y &= [x_- + y_-, x_+ + y_+], \\ XY &= [\min S, \max S]. \end{aligned}$$

Here $S = \{x_-y_-, x_-y_+, x_+y_-, x_+y_+\}$. Division by an interval Y not containing zero means multiplication by $[1/y_+, 1/y_-]$. All interval endpoints below are rational, and each square-root enclosure follows from explicit rational inequalities of the form

$$L^2 < q < U^2.$$

We prove Proposition 5.1 in four steps: uniform radical bounds, endpoint signs, strict monotonicity on J , and separation of the six roots.

A.1. Step 1: Uniform radical enclosures. We use the notation T_-, T_+, J fixed in Section 1. The derivatives P'_1, P'_2, P'_3 are increasing affine functions, and

$$P'_1(T_+) < 0, \quad P'_2(T_-) > 0, \quad P'_3(T_+) < 0.$$

Hence P_1 and P_3 are decreasing on J , while P_2 is increasing. The following uniform bounds hold:

$$\begin{aligned} \sqrt{11} &\in [3.31662, 3.31663], & \sqrt{P_1(T)} &\in [13.13216, 13.13243], \\ \sqrt{41} &\in [6.40312, 6.40313], & \sqrt{P_2(T)} &\in [78.22274, 78.22286], \\ \sqrt{65} &\in [8.06225, 8.06226], & \sqrt{P_3(T)} &\in [119.95844, 119.95962]. \end{aligned} \quad (47)$$

The six enclosures are established by the following positive rational margins:

q	$q_{\min} - L^2$	$U^2 - q_{\max}$
11	0.0000317756	0.0000345569
41	0.0000542656	0.0000737969
65	0.0001249375	0.0000363076
$P_1(J)$	0.0000557632	0.0002857049
$P_2(J)$	0.0001468924	0.0013700996
$P_3(J)$	0.0002301040	0.0023665444

Here $[L, U]$ is the corresponding interval in (47), while $q_{\min}$ and $q_{\max}$ are the appropriate monotone endpoint values on J . Positivity of the two displayed margins gives

$$L^2 < q_{\min} \leq q \leq q_{\max} < U^2,$$

and hence proves each enclosure in (47).

A.2. Step 2: The endpoint signs. At the two endpoints T_- and T_+ , the following sharper rational enclosures hold:

	$T = T_-$	$T = T_+$
$\sqrt{P_1}$	[13.132419, 13.132420]	[13.132162, 13.132163]
$\sqrt{P_2}$	[78.222740, 78.222741]	[78.222851, 78.222852]
$\sqrt{P_3}$	[119.959610, 119.959611]	[119.958440, 119.958441]

and

$$\begin{aligned} \sqrt{11} &\in [3.316624, 3.316625], & \sqrt{41} &\in [6.403124, 6.403125], \\ \sqrt{65} &\in [8.062257, 8.062258]. \end{aligned}$$

Each enclosure follows from the strict rational inequalities $L^2 < q < U^2$, with $q = P_\nu(T_\pm)$ or $q \in \{11, 41, 65\}$, as appropriate.

Let

$$a = \sqrt{11}, \quad b = \sqrt{41}, \quad c = \sqrt{65}, \quad s_\nu(T) = \sqrt{P_\nu(T)}.$$

The six brackets in (24), formed from the numerator lifts H_{ij} , are

$$\begin{aligned}
B_1 &= (57 + 25a)(s_2 + 600T - 281) - (115 + 5a)(s_2 + 351 - 360T), \\
B_2 &= (s_1 + 33 - 24T)(s_3 + 125 - 240T) \\
&\quad - (s_1 + 40T - 23)(576T + 5s_3 - 1077), \\
B_3 &= -(5 + b) + 4(c - 8), \\
B_4 &= -(57 + 25a) - (115 + 5a)(c - 8), \\
B_5 &= -4(s_1 + 33 - 24T) - (5 + b)(s_1 + 40T - 23), \\
B_6 &= (s_2 + 351 - 360T)(s_3 + 125 - 240T) \\
&\quad - (s_2 + 600T - 281)(576T + 5s_3 - 1077).
\end{aligned}$$

Substitution of the preceding rational enclosures into $B_1, \dots, B_6$ yields

	$T = T_-$	$T = T_+$
B_1	[24079.175326, 24079.183624]	[24080.489444, 24080.497742]
B_2	[1189.410506, 1189.410696]	[1189.296098, 1189.296288]
B_3	[-11.154097, -11.154092]	[-11.154097, -11.154092]
B_4	[-148.107728, -148.107570]	[-148.107728, -148.107570]
B_5	[-370.732120, -370.732080]	[-370.731762, -370.731723]
B_6	[5817.982138, 5817.983790]	[5817.378762, 5817.380415]

The two triple products themselves satisfy

	$B_1 B_2 B_3$	$B_4 B_5 B_6$
$T = T_-$	[-319453768, -319453463]	[319455087, 319455551]
$T = T_+$	[-319440473, -319440168]	[319421649, 319422113].

Since $\Delta_H = B_1 B_2 B_3 + B_4 B_5 B_6$, adding the two intervals in each row gives

$$\Delta_H(T_-) \in [1319, 2088] \subset (0, \infty), \quad (48)$$

$$\Delta_H(T_+) \in [-18824, -18055] \subset (-\infty, 0). \quad (49)$$

A.3. Step 3: Strict monotonicity on the isolating interval. Write $s_\nu = \sqrt{P_\nu}$, so $s'_\nu = P'_\nu / (2s_\nu)$. From (47) one obtains, throughout J ,

$$\begin{aligned}
s'_1 &\in [-25.701, -25.699], & s'_2 &\in [11.014, 11.047], \\
s'_3 &\in [-116.920, -116.916].
\end{aligned}$$

Differentiating the three moving lifts gives

$$\begin{aligned}
H'_{13} &= (s'_1 - 24, s'_1 + 40), \\
H'_{23} &= (s'_2 - 360, s'_2 + 600), \\
H'_{32} &= (576 + 5s'_3, s'_3 - 240).
\end{aligned}$$

The constant lifts have zero derivative. Hence

$$\begin{aligned}
B'_1 &= [H_{12}, H'_{23}], \\
B'_2 &= [H'_{13}, H_{32}] + [H_{13}, H'_{32}], \\
B'_5 &= [H'_{13}, H_{21}], \\
B'_6 &= [H'_{23}, H_{32}] + [H_{23}, H'_{32}]
\end{aligned}$$

and $B'_3 = B'_4 = 0$. The uniform radical bounds (47) give, throughout J ,

$$\begin{array}{l|l} B_1 & [24079, 24081] \\ B_2 & [1189, 1190] \\ B_3 & [-11.155, -11.154] \\ B_4 & [-148.109, -148.106] \\ B_5 & [-370.738, -370.726] \\ B_6 & [5815, 5821] \end{array} \quad \begin{array}{l|l} B'_1 & [131406, 131416] \\ B'_2 & [-11449, -11447] \\ B'_5 & [35, 36] \\ B'_6 & [-60459.78, -60442.07] \end{array}$$

Since $B'_3 = B'_4 = 0$, differentiation gives

$$\Delta'_H = B'_1 B_2 B_3 + B_1 B'_2 B_3 + B_4 B'_5 B_6 + B_4 B_5 B'_6.$$

The displayed bounds give

$$B'_1 B_2 B_3 \in [-1744475122, -1742720301],$$

$$B_1 B'_2 B_3 \in [3074402819, 3075471082],$$

$$B_4 B'_5 B_6 \in [-31037130, -30143273],$$

$$B_4 B_5 B'_6 \in [-3319824419, -3318677322].$$

Adding these four rational intervals gives

$$\Delta'_H(T) \in [-2020933852, -2016069814] \subset (-\infty, 0), \quad T \in J. \quad (50)$$

A.4. Step 4: Separation of the six roots. Applying the uniform bounds (47) to the fifteen pairwise brackets yields the following strict estimates. The row and column labels denote the ordered pairs indexing the lifts; for example, the entry in row 12, column 13 means $[H_{12}, H_{13}] < -188$ throughout J .

	13	21	23	31	32
12	< -188	< -2060	> 24079	< -148	> 6340
13		< -370	> 4689	< -27	> 1189
21			> 3885	< -11	> 525
23				< -148	> 5815
31					> 3

Every bracket is therefore nonzero on J . The endpoint signs (48), the derivative bound (50), and the fifteen nonvanishing estimates complete the proof of Proposition 5.1.

REFERENCES

- [1] M. F. Atiyah, *The geometry of classical particles*, in *Surveys in Differential Geometry*, Surv. Differ. Geom. **VII**, International Press, Somerville, MA, 2000, 1–15, MR 1919420.
- [2] M. F. Atiyah, *Configurations of points*, Philos. Trans. Roy. Soc. London Ser. A **359** (2001), no. 1784, 1375–1387, doi:10.1098/rsta.2001.0840.
- [3] M. F. Atiyah, *Edinburgh lectures on geometry, analysis and physics*, notes by T. Köppe, preprint (2010), arXiv:1009.4827 [math.AG].
- [4] M. F. Atiyah and R. Bielawski, *Nahm's equations, configuration spaces and flag manifolds*, Bull. Braz. Math. Soc. (N.S.) **33** (2002), no. 2, 157–176, doi:10.1007/s005740200007.
- [5] M. F. Atiyah and P. M. Sutcliffe, *The geometry of point particles*, Proc. Roy. Soc. London Ser. A **458** (2002), no. 2021, 1089–1115, doi:10.1098/rspa.2001.0913.
- [6] M. V. Berry and J. M. Robbins, *Indistinguishability for quantum particles: spin, statistics and the geometric phase*, Proc. Roy. Soc. London Ser. A **453** (1997), no. 1963, 1771–1790, doi:10.1098/rspa.1997.0096.
- [7] M. N. Bou Khuzam and M. J. Johnson, *On the conjectures regarding the 4-point Atiyah determinant*, SIGMA Symmetry Integrability Geom. Methods Appl. **10** (2014), Paper 070, 9 pp., doi:10.3842/SIGMA.2014.070.
- [8] D. Z. Đoković, *Proof of Atiyah's conjecture for two special types of configurations*, Electron. J. Linear Algebra **9** (2002), 132–137, doi:10.13001/1081-3810.1081.
- [9] D. Z. Đoković, *Verification of Atiyah's conjecture for some nonplanar configurations with dihedral symmetry*, Publ. Inst. Math. (Beograd) (N.S.) **72(86)** (2002), 23–28, doi:10.2298/PIM0272023D.

- [10] M. Eastwood and P. Norbury, *A proof of Atiyah's conjecture on configurations of four points in Euclidean three-space*, *Geom. Topol.* **5** (2001), 885–893, doi:10.2140/gt.2001.5.885.
- [11] L. Guerra and P. Salvatore, *The Atiyah–Sutcliffe conjecture and E_n -algebras*, preprint (2024), arXiv:2410.24124 [math.AT].
- [12] J. Malkoun, *Configurations of points and the symplectic Berry–Robbins problem*, *SIGMA Symmetry Integrability Geom. Methods Appl.* **10** (2014), Paper 112, 6 pp., doi:10.3842/SIGMA.2014.112.
- [13] J. Malkoun, *On the Atiyah problem on hyperbolic configurations of four points*, *Geom. Dedicata* **180** (2016), 287–292, doi:10.1007/s10711-015-0102-8.
- [14] J. Malkoun, *Constructions of $SU(2)$ and Weyl equivariant maps for all classical groups*, preprint (2015, revised 2021), arXiv:1508.04076 [math.MG].
- [15] J. Malkoun, *Configuration spaces of points, symmetric groups and polynomials of several variables*, preprint (2015), arXiv:1509.06629 [math.MG].
- [16] J. Malkoun, *Towards the Atiyah–Sutcliffe conjectures for coplanar hyperbolic points*, preprint (2019), arXiv:1909.00571 [math.MG].
- [17] J. Malkoun, *A new proof of Atiyah's conjecture on configurations of four points*, *J. Math. Phys.* **64** (2023), no. 5, Paper 053507, doi:10.1063/5.0151938.
- [18] J. Malkoun, *Root systems and the Atiyah–Sutcliffe problem*, *J. Math. Phys.* **60** (2019), no. 10, Paper 101702, doi:10.1063/1.5111442.
- [19] J. Malkoun, *Weights, Weyl-equivariant maps and a rank conjecture*, *Exp. Math.* **31** (2022), no. 3, 843–846, doi:10.1080/10586458.2020.1712272.
- [20] J. Malkoun, *The Atiyah–Sutcliffe determinant*, preprint (2019), arXiv:1903.05957 [math.MG].
- [21] J. Malkoun, *Finite graphs and configurations of points*, preprint (2025, revised 2026), arXiv:2508.13472 [math.CO].
- [22] M. Mazur and B. V. Petrenko, *On the conjectures of Atiyah and Sutcliffe*, *Geom. Dedicata* **158** (2012), 329–342, doi:10.1007/s10711-011-9636-6.
- [23] D. Svrtan and I. Urbiha, *Atiyah–Sutcliffe conjectures for almost collinear configurations and some new conjectures for symmetric functions*, preprint (2004), arXiv:math/0406386 [math.AG].
- [24] D. Svrtan and I. Urbiha, *Verification and strengthening of the Atiyah–Sutcliffe conjectures for several types of configurations*, preprint (2006), arXiv:math/0609174 [math.MG].

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